Dr. Koushik Bera

PhD in Statistics · IIT Guwahati · Assam (India) · 781039

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Research Interests

Time Series Analysis, Financial Econometrics, Statistical Machine Learning, Volatility Modelling, Risk Forecasting, High-Frequency Financial Data Analysis, Portfolio Optimisation.

Education

Ph.D. in Statistics *2021 - 2026*
Department of Mathematics, Indian Institute of Technology Guwahati · CGPA: 9.02

M.Sc. in Mathematics *2019 - 2021*
Department of Mathematics, Indian Institute of Technology Bhubaneswar · CGPA: 9.06

B.Sc. (Hons.) in Mathematics *2016 - 2019*
Midnapore College (Autonomous), Vidyasagar University · 79.38%

Higher Secondary *2016*
WBCHSE Board · 92.00%

Secondary *2014*
WBBSE Board · 84.71%

Ph.D. Details

- **Thesis Title:** Statistical and Deep Learning Approaches to Volatility Forecasting in Financial Time Series across Diverse Financial Markets.
- **Supervisor:** Prof. Selvaraju Natarajan
- **Institute:** Indian Institute of Technology Guwahati, Assam, India
- **Status:** Defended on 30th June 2026.

Publications

Journal Articles:

1. **Koushik Bera**, N. Selvaraju. A leakage-free optimized Variational Mode Decomposition-based state space framework for crude oil volatility forecasting, *Energy and AI*, 25, 100781, 2026. DOI: 10.1016/j.egyai.2026.100781
2. **Koushik Bera**, A. Bhambu, N. Selvaraju. A Heteroscedasticity-based Extended Long Short Term Memory Network for Volatility Forecasting and Risk Estimation, *Applied Soft Computing*, 115627, 2026. DOI: 10.1016/j.asoc.2026.115627
3. **Koushik Bera**, N. Selvaraju. Extended-Horizon Realized Volatility Forecasting Using State Space Models, *Systems and Soft Computing*, 7, 200427, 2025. DOI: 10.1016/j.sasc.2025.200427
4. **Koushik Bera**, P. Raj, N. Selvaraju. An Integrated Framework for Volatility Prediction: Leveraging Decomposition Techniques with Realized GARCH Models, *Computational Economics*, 67, 3275–3309, 2026. DOI: 10.1007/s10614-025-10930-3
5. **Koushik Bera**, P. Raj, N. Selvaraju. Forecasting Financial Volatility: A Hybrid Approach Integrating ICEEMDAN and Asymmetric Volatility Models, *Journal of Financial Data Science*, 7(1), pp 54-75, 2025. DOI: 10.3905/jfds.2024.1.174
6. A. Maity, **Koushik Bera**, N. Selvaraju. Behavioral robust mean-variance portfolio selection with an intractable claim, *Mathematics and Financial Economics*, pp 1-29, 2025. DOI: 10.1007/s11579-025-00386-2
7. A. Bhambu, **Koushik Bera**, N. Selvaraju, Ponnuthurai Nagaratnam Suganthan. High frequency volatility forecasting and risk assessment using neural networks-based heteroscedasticity model, *Engineering Applications of Artificial Intelligence*, 149, pp 110397, 2025. DOI: 10.1016/j.engappai.2025.110397
8. P. Raj, **Koushik Bera**, N. Selvaraju. A hybrid model for intraday volatility prediction in Bitcoin markets, *The North American Journal of Economics and Finance*, 78, 2025, 102426. DOI: 10.1016/j.najef.2025.102426

9. P. Raj, **Koushik Bera**, N. Selvaraju. Power of Decomposition in Volatility Forecasting for Bitcoins, *Pacific-Basin Finance Journal*, 93, 102839, 2025. DOI: 10.1016/j.pacfin.2025.102839
10. P. Raj, **Koushik Bera**, N. Selvaraju. Do returns still matter? A complete asymmetric volatility model with realized measures in financial markets, *Finance Research Open*, 2, 100139, 2026. DOI: 10.1016/j.finr.2026.100139

Conference Proceedings:

1. **Koushik Bera**, A. Bhambu, P. Raj, N. Selvaraju. A Heteroscedastic Neural Network Framework for Volatility Prediction, *ICOAISC-2024*, NIT Nagaland, 2024. DOI: 10.37285/bsp.oaisem2025.09
2. A. Bhambu, **Koushik Bera**, P. Raj, N. Selvaraju. Deep Learning Based Decomposition Methods for Financial Volatility Forecasting, Current Progress in Interdisciplinary Research; *RIC*, Volume 3, Springer, 2025. DOI: 10.1007/978-981-95-1853-1_21

Communicated Articles:

1. P. Raj, A. Bhambu, **Koushik Bera**, N. Selvaraju. Volatility Forecasting: A Bootstrap-Enhanced Deep Neural Network Approach. (Communicated)
2. **Koushik Bera**, N. Selvaraju. Volatility-Driven Portfolio Optimization with Advanced Volatility Forecasting Models. (Communicated) .
3. P. Raj, **Koushik Bera**, A. Bhambu, N. Selvaraju, A. Sinha. Does the Distribution of Financial Returns Matter? Insights from Fat-Tailed Models in Volatility Forecasting. (Communicated)

Projects

- **A Review of Prediction of Rainfall Using A Kalman Filtering Algorithm Approach** *June 2020 - June 2021*
  Dr. Abhijit Datta Banik, IIT Bhubaneswar
  - Utilized multivariate regression, AR model, PCA, and Kalman Filtering algorithm to predict rainfall.

- **Weather Prediction Using Machine Learning.** *CSIR-SRTP 2020*
  Dr. Naresh Kumar, CSIR-NEIST
  - Explored machine learning techniques for weather prediction as part of the CSIR-Summer Research Training Program 2020 Online.

Technical Skills

1. Programming: Python, C, R

Teaching Assistantship

•MA 225: Probability Theory & Random Process	IIT Guwahati	*Jul 23 – Nov 25 (3 semesters)*
•DA 210: Time Series Analysis & Forecasting	Coursera	*May 25 – Aug 25*
•MA 374: Financial Engineering Laboratory	IIT Guwahati	*Jan 24 – May 25 (2 semesters)*
•MOOC: Introduction to Queueing Theory	NPTEL	*Jan 24 – Jul 25 (2 semesters)*
•DA 206: Statistical Inferencing	Coursera	*Dec 24 – Jun 25*
•MA 101: Real Analysis, Single and Multivariable Calculus	IIT Guwahati	*Nov 22 – Mar 23*

Positions of Responsibility

- **Member:** Research Scholars' Forum Mathematics *2023 - 2024*
- **Reviewer:** Engineering Applications of Artificial Intelligence, Scientific Reports, Humanities & Social Sciences Communications *Ongoing*

Achievements

- **Joint CSIR-UGC NET:** JRF, AIR-144 *June - 2021*
- **GATE Mathematics:** AIR-568 *2021*
- **GATE Statistics:** AIR-96 *2023*
- **IIT JAM Mathematics:** AIR-326 *2019*
- **IIT JAM Statistics:** AIR-274 *2019*
- **Inspire Scholarship:** Innovation in Science Pursuit for Inspired Research Scholarship *2016-2021*
- **Best Thesis Award:** MSc Project, IIT Bhubaneswar *2020-2021*
- **Best Poster Presentation Award:** RTQF 2026, IISc Bangalore *February - 2026*

Workshops/Conferences

- Recent Trends in Quantitative Finance 2026 (RTQF 2026) IISc Bangalore, February 20-21, 2026
- The Fifth Indian Symposium on Machine Learning (IndoML) 2024 BITS Pilani Goa Campus, Goa, December 21-23, 2024
- International Conference On Optimization with Artificial Intelligence and Soft-Computing, 2024 NIT Nagaland, Nagaland, 30 September - 01 October, 2024. Presented a paper titled "A Heteroscedastic Neural Network Framework for Volatility Prediction"
- International Workshop and Conference on 'Statistical Methods in Finance', 2023 Chennai Mathematical Institute, Chennai, December 19-23, 2023
- Bombay Workshop on Statistics and Probability IIT Bombay, January 3-6, 2023
- Instructional School for Teachers (Under ATM Schools), Advanced Topics in Statistical Learnin Indian Statistical Institute, Kolkata, July 11-23, 2022